

HARMAN AUJLA

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EDUCATION

Rensselaer Polytechnic Institute

Bachelor of Science in Computer Science and Economics

Expected Graduation: May 2027

Troy, NY

- GPA: 3.6
- Relevant Coursework: Data Structures, Algorithms, Computer Organization, Machine Learning from Data, Reinforcement Learning, Data Science, Computer Vision, Projects in AI/ML, Machine Learning and Optimization, Database Systems, Econometrics

EXPERIENCE

TD Securities

June 2025 – August 2025

Software Engineering Intern

New York, NY

- Built an **LLM**-assisted log analysis system using Python, **PyTorch**, and **Hugging Face Transformers** to embed and analyze 10K+ **Jenkins** build logs across 15+ CI pipelines, identifying recurring failure patterns and reducing incidents by **~20%**
- Designed and deployed **Java Spring Boot** microservices supporting **10K+** daily payment transactions, achieving <250ms average latency and **~90%** unit test coverage in a regulated financial environment
- Developed and maintained production-grade **CI/CD pipelines** using **Jenkins**, **Shell scripting**, **Veracode**, and **SonarQube**, automating testing, security scanning, and deployment; supported large-scale **VMware Cloud** data migration and validation

International Socioeconomic Laboratory

June 2023 – August 2023

Research Intern

New York, NY

- Trained and evaluated a **TensorFlow**-based neural network classifier to model sector preference patterns and test whether demographic features provided predictive signal beyond classical statistical methods
- Co-authored a **research paper** submitted to a peer-reviewed journal analyzing public preferences for sector-specific tax aid during the COVID-19 pandemic
- Conducted statistical and **machine learning** analysis on survey data using **R**, **RStudio**, and **NumPy**, including multivariate regression, hypothesis testing, and robustness checks

RPI AI Club

Troy, NY

AI Developer

August 2024 – Present

- Developed **RPI GPT**, a campus-wide AI platform providing a Pocket TA for courses using **RAG pipelines** and **fine-tuned LLMs** trained on instructor-uploaded materials for course-specific student assistance
- Designed a scalable backend AI pipeline integrating document ingestion, embedding generation, vector search, and response orchestration, with a **React** frontend for real-time student interaction
- Built and evaluated **neural network models** using **PyTorch**, including an MLP on the **MNIST** dataset and a **CNN on CIFAR-10**, to benchmark training workflows, optimization strategies, and model performance

PROJECTS

SYEP Management System | *Next.js, React, Node.js, PostgreSQL, Docker*

June 2024 – August 2024

- Developed and deployed a **full-stack** maintenance management platform enabling custodial staff and supervisors to log, track, and audit daily and monthly facilities tasks, with the application containerized using Docker for consistent deployment

FinLLM Web Search Agent | *PyTorch, Hugging Face, Next.js, React*

August 2024 – December 2024

- Developed an **AI**-driven web search and financial advisory platform combining real-time **web scraping**, **RAG**, and structured database integration to provide accurate financial insights; enhanced LLM performance through fine-tuning, benchmarking, and parsing of local financial documents to support reliable decision-making.

TECHNICAL SKILLS

Languages/Tools: Java, Python, C/C++, SQL, JavaScript, Node.js, Swift, React, Git, Docker, HTML/CSS, R
ML Skills: NN, CNN, RNN, LSTM, Trees, Q Learning, NLP, AEs, VAEs, Transformers, GANs, Diffusion Models, V-JEPA

Honors: RPI Hackathon Winner, Dean's Honor Roll